

Name: Srinjoy Chakraborty

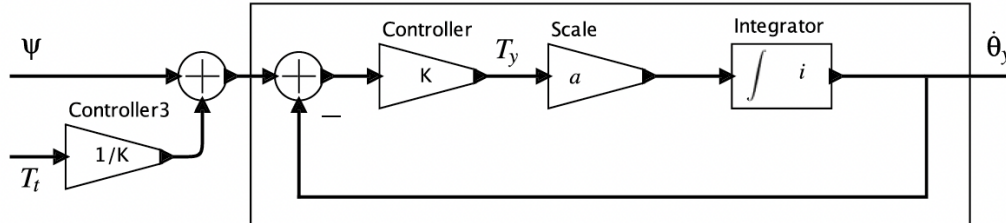
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Homework Assignment 1

Deadline: September 11, 2026

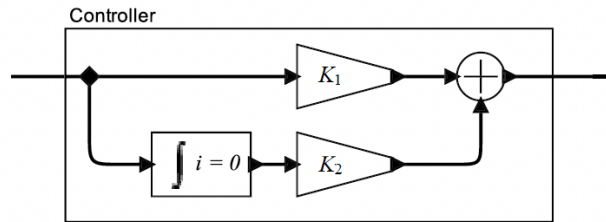
Total: 60 marks**Problem 1.** (10 points)

(a) Construct a Simulink model of the helicopter control system shown in the figure below.



Choose some reasonable parameters and plot the actual angular velocity as a function of time, assuming that the desired angular velocity is zero, $\psi(t) = 0$, and that the top-rotor torque is non-zero, $T_t(t) = bu(t)$. Give your plot for three different values of K and discuss how the behavior varies with K .

(b) Modify the model of part (a) to replace the Controller (the simple scale-by- K actor) with the alternative controller shown in the figure below.



This alternative controller is called a proportional-integrator (PI) controller. It has two parameters K_1 and K_2 . Experiment with the values of these parameters, give some plots of the behavior with the same inputs as in part (a), and discuss the behavior of this controller in contrast to the one of part (a).

(a) Proportional Controller Behavior

I built a Simulink model to test the proportional scale-by- K controller. I ran three systems in parallel so I could compare different values of K at the same time. The desired angular velocity was kept at $\psi(t) = 0$, and I used a unit step input to model the top-rotor torque $T_t(t)$. The parameters I picked are:

- **Case 1:** $K = 4$, Scale $a = 2$, Torque Input Gain = $1/4$
- **Case 2:** $K = 5$, Scale $a = 2$, Torque Input Gain = $1/5$
- **Case 3:** $K = 6$, Scale $a = 2$, Torque Input Gain = $1/6$

Discussion: Since there is a single integrator in the system, the closed-loop response behaves like a first-order system. Hence, the response to a torque disturbance appears as a smooth exponential decay curve, with no overshoot. In steady state, the angular velocity eventually settles at $1/K$ in steady state. So basically, bigger K results in a smaller steady-state error and faster settling time, as the time constant also decreases.

Name: Srinjoy Chakraborty

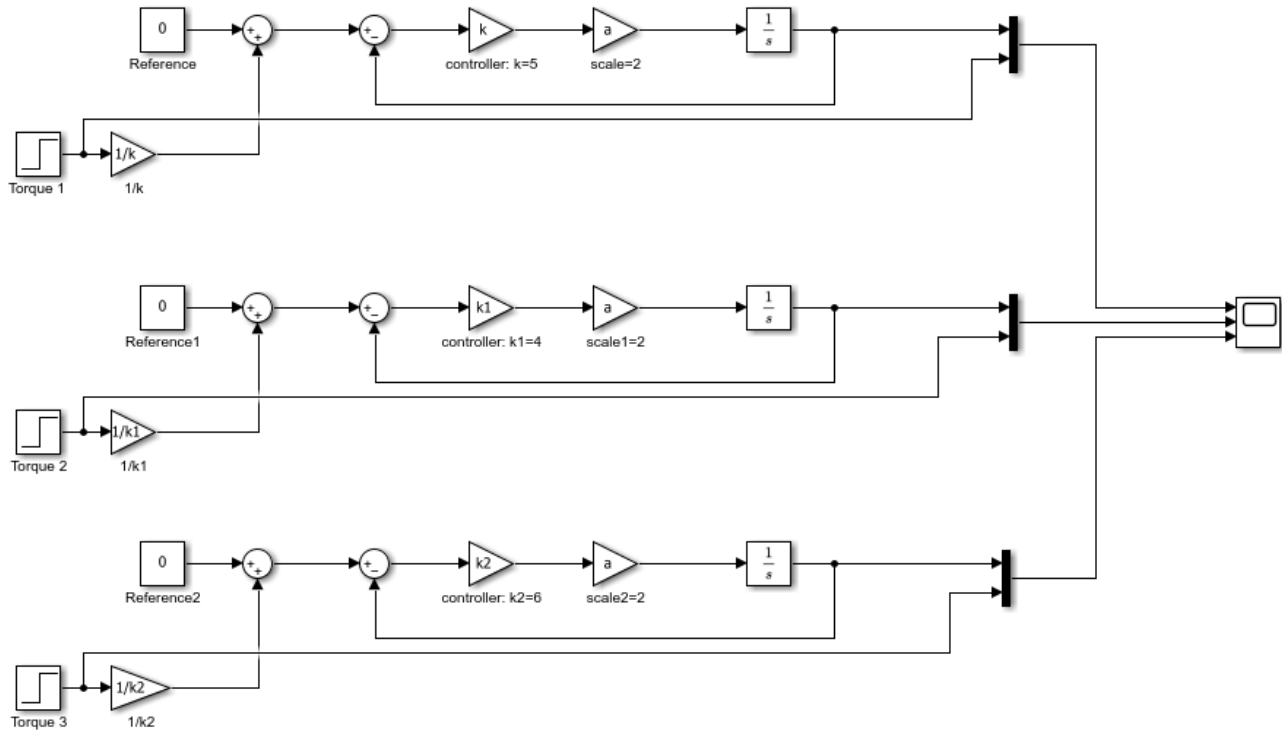
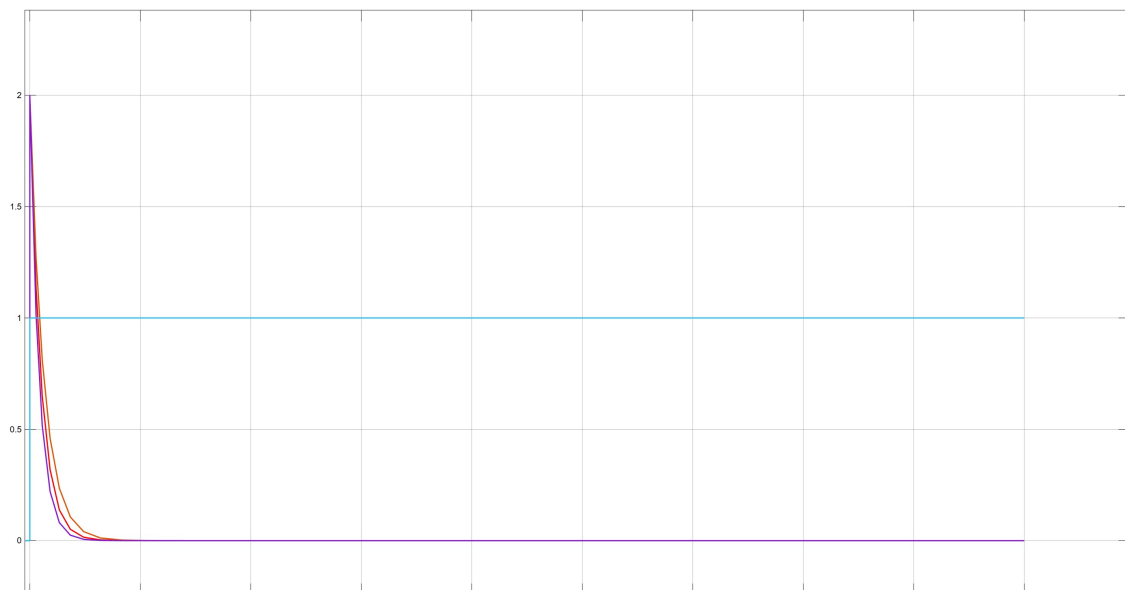
Indian Institute of Technology Kanpur
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e.g. CSEHomework Assignment 1
Deadline: September 11, 2026Figure 1.1: Simulink layout for Part (a) testing three different K values in parallel.

Figure 1.2: Error over time for Part (a).

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

(b) Proportional-Integrator (PI) Controller Behavior

Here I swapped out the simple proportional controller for a PI controller that has two gains, K_1 (proportional) and K_2 (integral). I also updated the torque input gain to $1/K_1$ to keep the disturbance scaling consistent. I tested three cases in parallel again:

- **Case 1 (Top Loop):** $K_1 = 4$, $K_2 = 2$, Scale $a = 2$, Torque Gain = $1/4$
- **Case 2 (Middle Loop):** $K_1 = 2$, $K_2 = 4$, Scale $a = 2$, Torque Gain = $1/2$
- **Case 3 (Bottom Loop):** $K_1 = 5$, $K_2 = 10$, Scale $a = 2$, Torque Gain = $1/5$

Adding the second integral term turns the closed loop into a second order system. Here now instead of an exponential decay, the response now overshoots and dips below the zero mark, before settling down finally. The integrator whereas makes sure the final steady state value lands exactly at $1/K_1$. Also I noticed that larger K_2 makes the oscillations more aggressive.

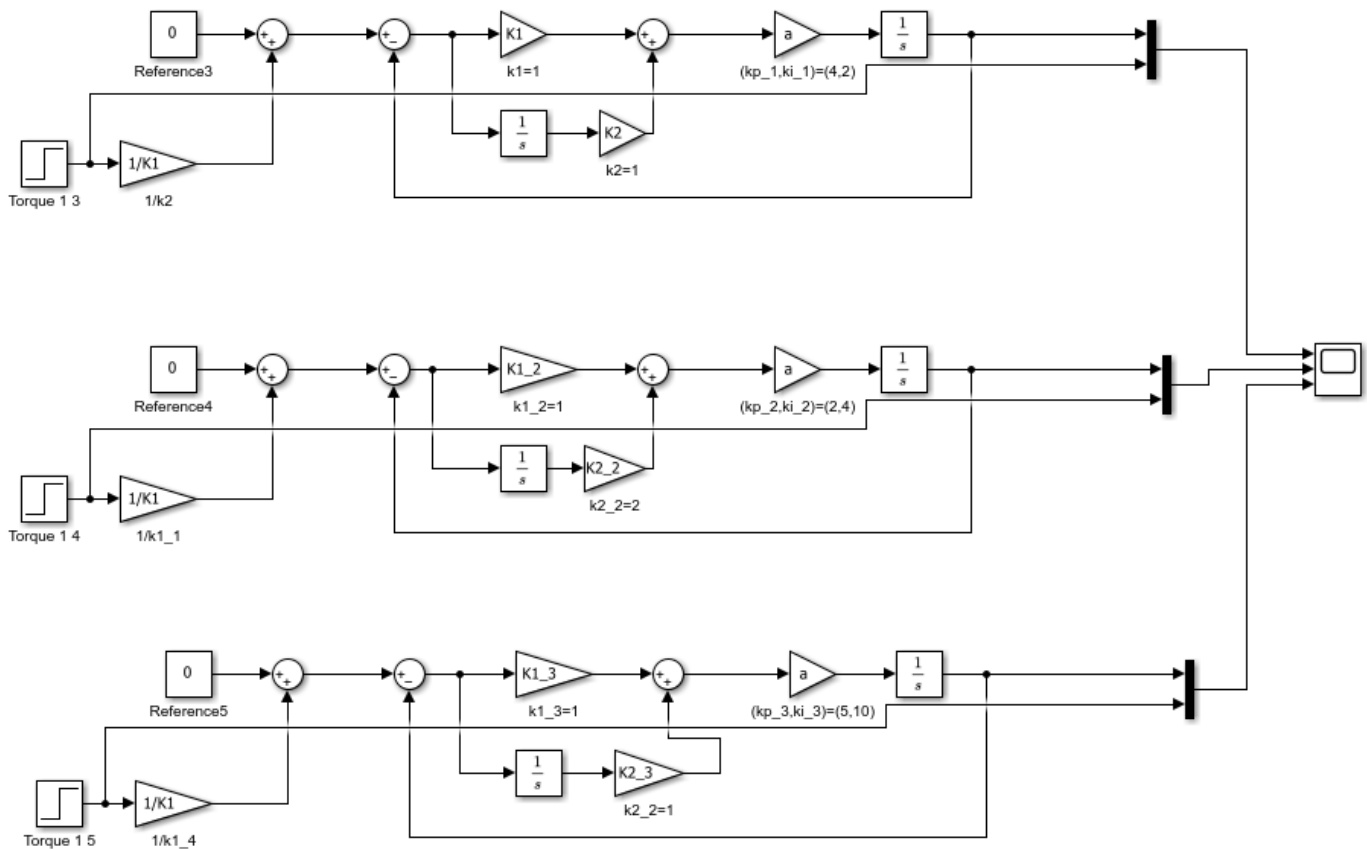


Figure 1.3: Modified Simulink layout of PI controller.

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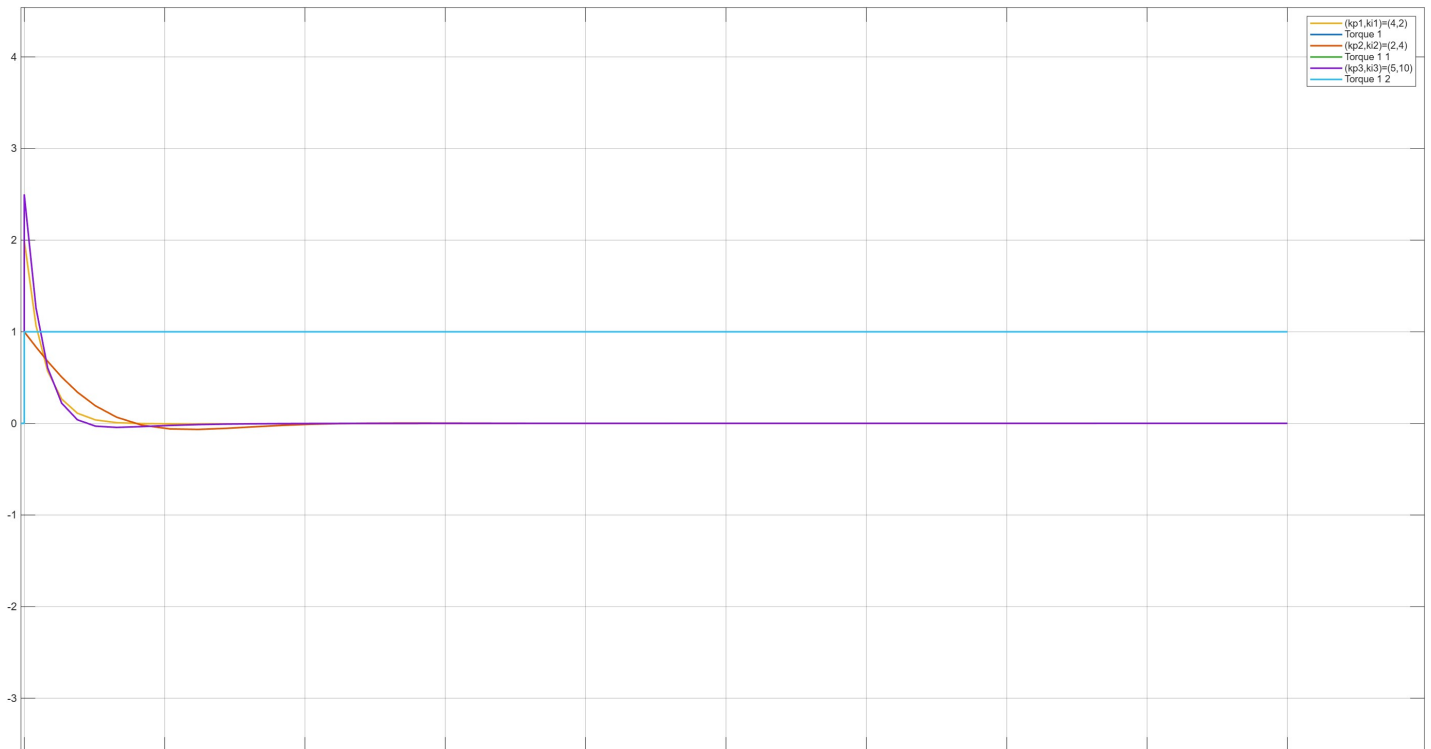
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e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 1.4: Error over time for Part (b).

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
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e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

Problem 2. (10 points)

The states of the linearized model of a vehicle steering system represent the lateral deviation of the vehicle from the x -axis and the angle between the vehicle axis and the x -axis. The output of the linearized model is only the first state. Construct a Simulink model for the vehicle steering system with its controller that includes an observer. The dynamics are available in Example 6.4 and Example 7.3 in [AM09]. Apply a sinusoidal signal as the reference trajectory that specifies the desired deviation of the vehicle from the x -axis with time. Plot the output (lateral deviation of the vehicle from the x -axis) with time.

[AM09] K. J. Astrom and R. M. Murray. Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press, 2009.

http://www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_22Feb09.pdf.

Using the normalized kinematic bicycle model from [AM09], the plant matrices for the lateral deviation system in continuous time are:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} \gamma \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0$$

To track a reference r and obtain the desired characteristic polynomial $p(s) = s^2 + 2\zeta_c\omega_c s + \omega_c^2$, the control law $u = -Kx + k_r r$ is used. By matching the closed-loop characteristic polynomial with the desired one, the feedback gains are obtained as

$$k_1 = \omega_c^2, \quad k_2 = 2\zeta_c\omega_c - \gamma\omega_c^2$$

For unit DC gain and proper steady-state tracking, the feedforward gain is chosen as $k_r = k_1 = \omega_c^2$. With $\omega_c = 1$ rad/s, $\zeta_c = 0.707$, and $\gamma = 0.5$, the resulting values are

$$K = \begin{bmatrix} 1 & 0.914 \end{bmatrix}, \quad k_r = 1.$$

Since only one state is measured, I needed an observer to estimate both states. Hence we design it to be twice as fast as the controller, so $\omega_o = 2\omega_c = 2$ rad/s with $\zeta_o = 0.707$. Matching $\det(sI - A + LC) = s^2 + l_1 s + l_2$ gives:

$$l_1 = 2\zeta_o\omega_o = 2.828, \quad l_2 = \omega_o^2 = 4 \implies L = \begin{bmatrix} 2.828 \\ 4 \end{bmatrix}$$

To set this up in Simulink using a State-Space block, it is defined the observer matrices as: $A_{obs} = A - LC$, $B_{obs} = \begin{bmatrix} B & L \end{bmatrix}$, and $C_{obs} = I_{2 \times 2}$ so it outputs the full estimated state $\hat{x}$ which then feeds into the gain K .

Name: Srinjoy Chakraborty

Roll No: 251040638
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CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

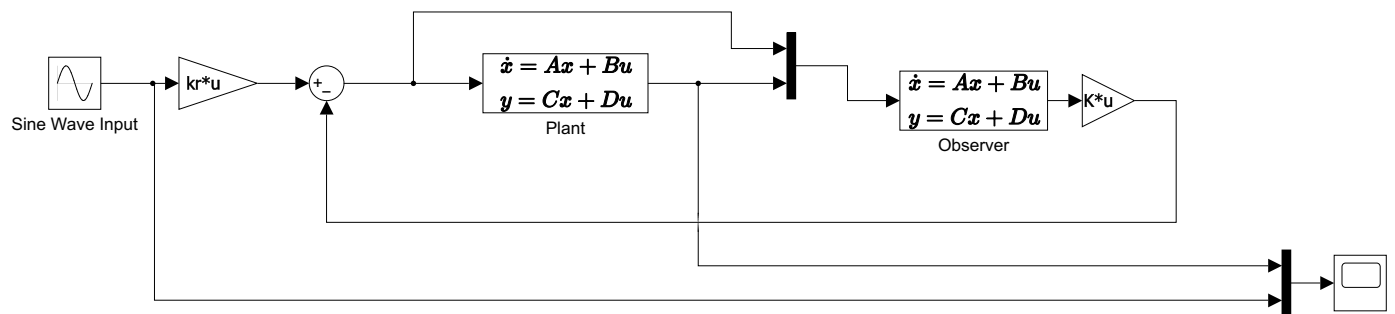


Figure 2.1: Simulink layout of the vehicle steering system.

Name: Srinjoy Chakraborty

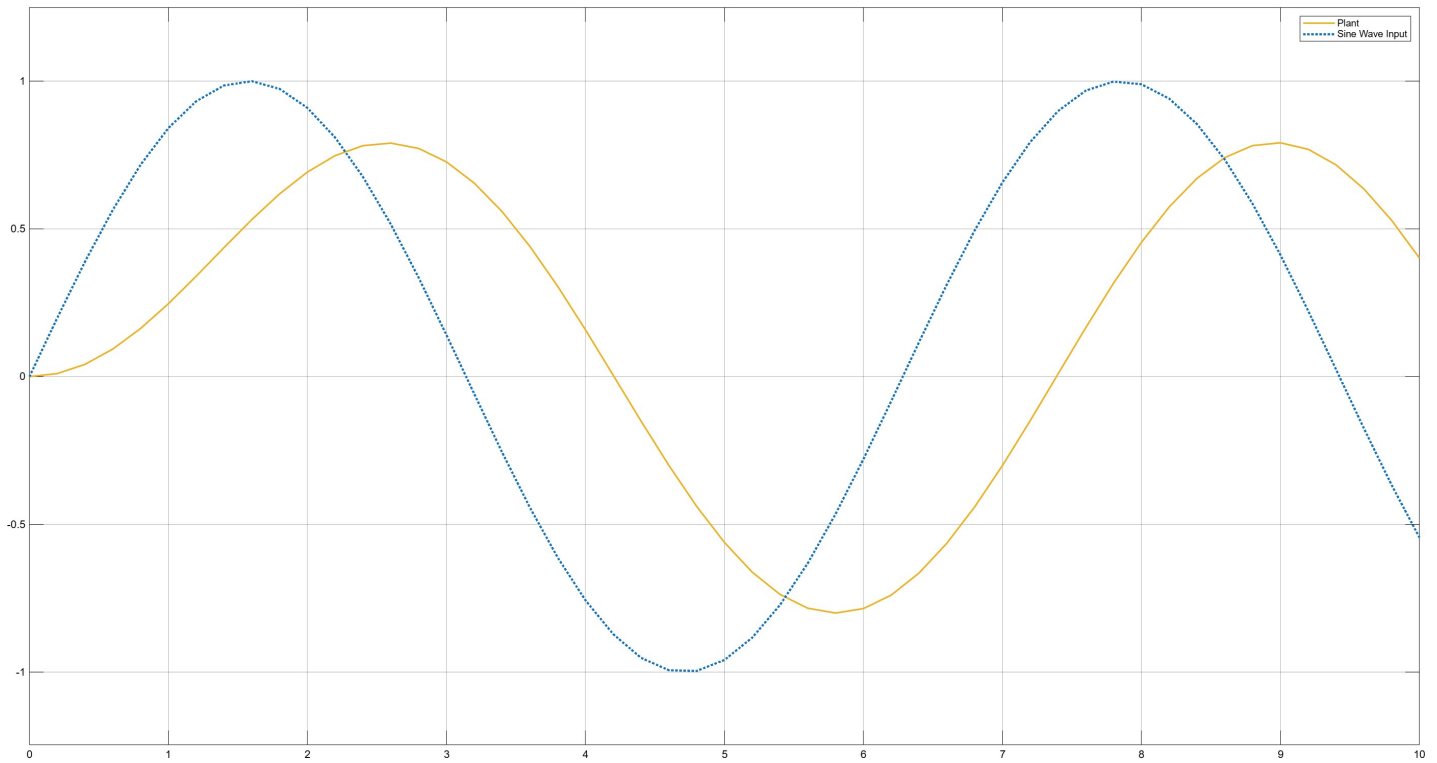
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e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 2.2: System response plotting the actual lateral deviation y tracking the sinusoidal reference trajectory using the observer.

Name: Srinjoy Chakraborty

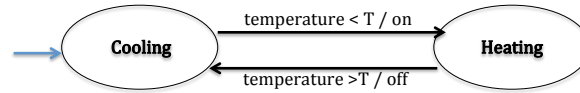
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CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

Problem 3. (10 points)

Consider the following model for a thermostat system.



The thermostat has been designed to maintain the temperature of a room at $T^{\circ}\text{C}$. The model has two states: *cooling* and *heating*. When the system is in the *cooling* state and the temperature of the room goes below $T^{\circ}\text{C}$, the system generates a signal to switch on a heater and moves to the *heating* state. When the temperature of the room goes over $T^{\circ}\text{C}$, the system generates a signal to switch off the heater and moves to the *cooling* state.

- Represent the system as an actor that takes the current temperature as input and produces a signal to control the heater. The actor uses the set point T as a parameter.
- Identify a design problem in the model.
- Provide two different remedies to address the problem.
- Compare your proposed two solutions in terms of ease of implementation and guarantees on the system behavior.

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSE

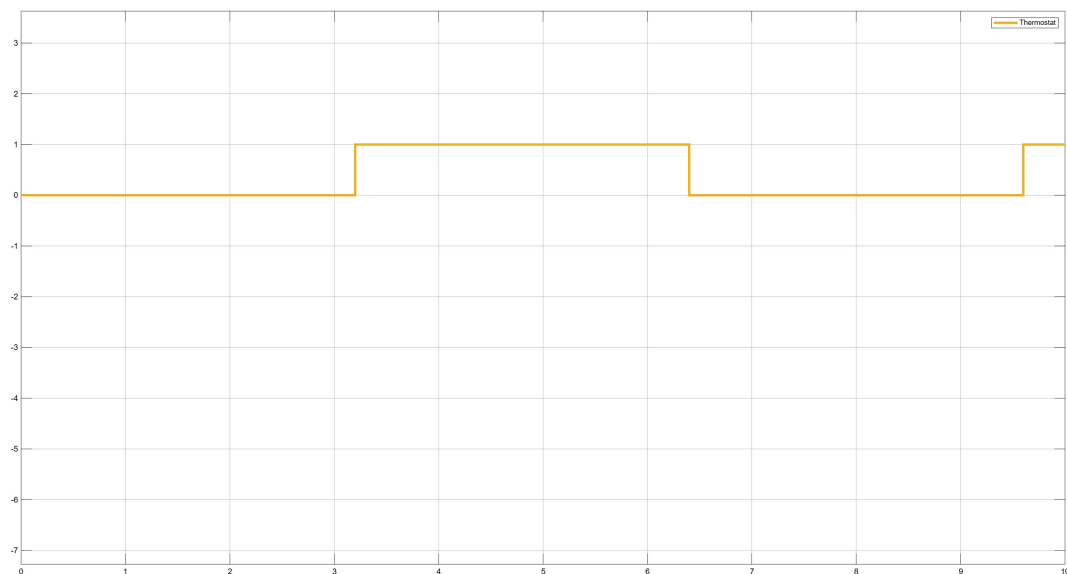
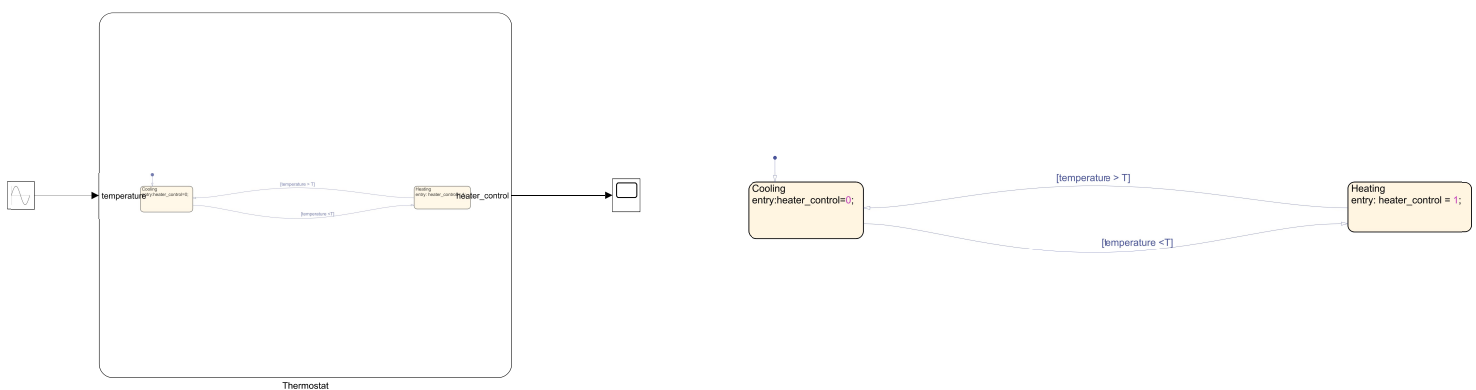
Homework Assignment 1

Deadline: September 11, 2026

I modeled the thermostat as a Stateflow chart in Simulink. It takes `temperature` as a continuous input and puts out `heater_control` as a discrete signal.

Initializations: The set point T is defined in the MATLAB workspace with parameter $T = 24$.

Connections: On the canvas, I connected a Sine Wave block to the `temperature` input to fake room temperature fluctuations with settings: **Amplitude=5**, **Bias=24**, **Frequency=1**. The `heater_control` output goes to a Scope along with the temperature signal.



Design Problem

The main problem here is **chattering**, which is also called Zeno behavior. Since the model has only one exact threshold at $T^\circ\text{C}$ and there is no buffer around it, even small sensor noise or minor temperature changes near T can make the system switch between the two states very quickly. In a real heater, this continuous switching can damage the physical relay and reduce its life quite fast.

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

Roll No: 251040638
e.g. 170001Dept.: EE
e.g. CSE

Remedies

Two ways to fix the chattering problem:

1. **Hysteresis:** Adding an error e around the set point of $T = 24^{\circ}\text{C}$. The heater turns on at $T - e$ and turns off at $T + e$.
2. **Temporal Delay:** Using a timer with temporal logic to make the system stay in whatever state it switches to for a fixed minimum time before it's allowed to switch again, no matter what the temperature is doing.

Comparison of Solutions

Ease of Implementation: Hysteresis is easier to implement, we just change the guard conditions on the transition arrows (e.g., `[temperature < T]` becomes `[temperature < T-e]`).

Guarantees on System Behavior: Hysteresis protects against noise as long as it stays within the $e = 2^{\circ}\text{C}$ band, but if the temperature swings wildly, rapid switching could still occur.

Name: Srinjoy Chakraborty

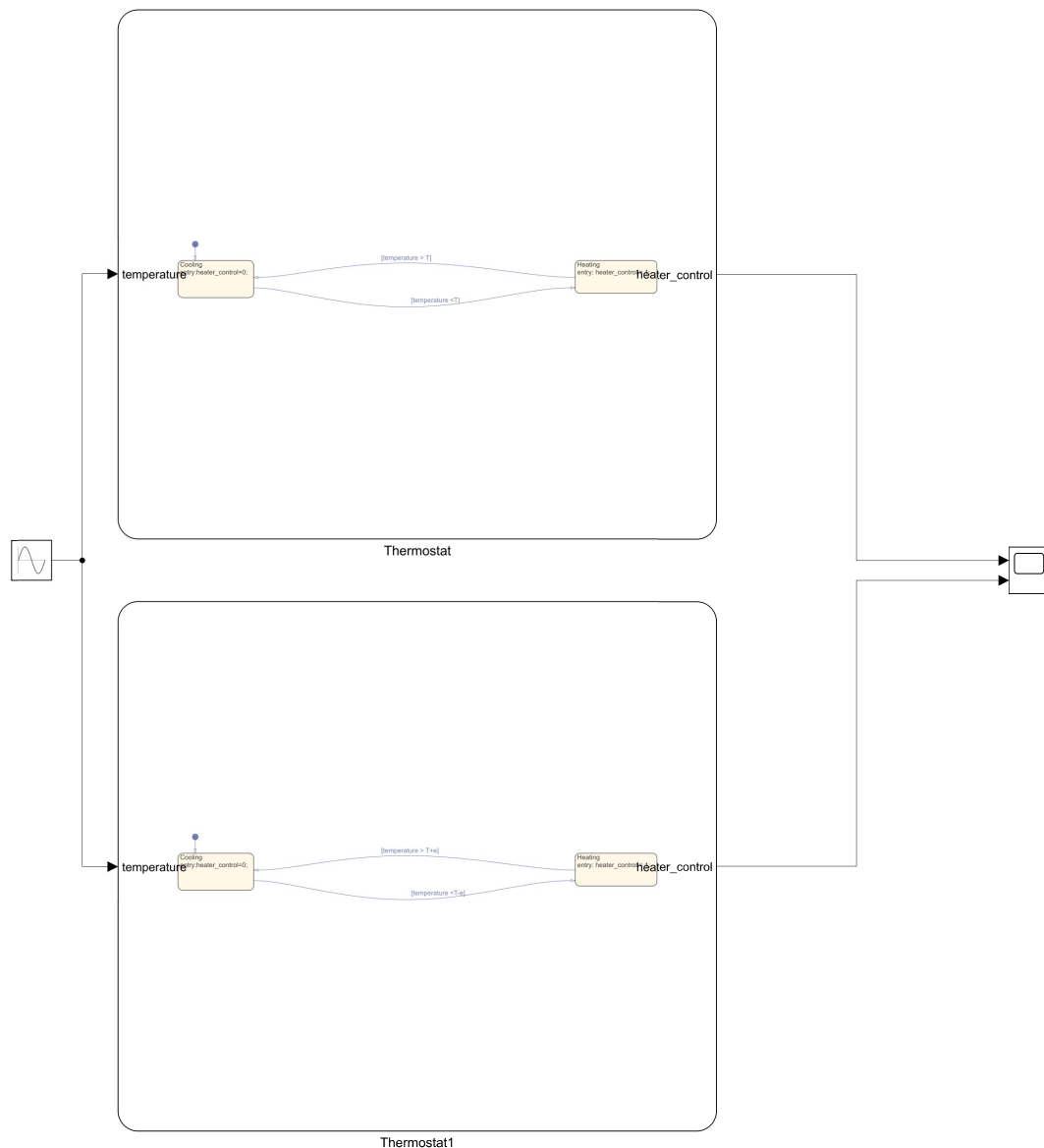
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CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 3.1: Simulink layout for the modified thermostat model.

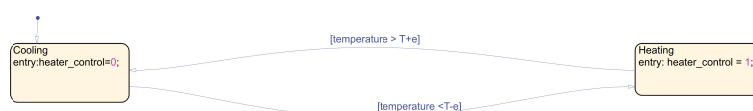


Figure 3.2: Stateflow chart demonstrating the hysteresis

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Roll No:

251040638

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Dept.:

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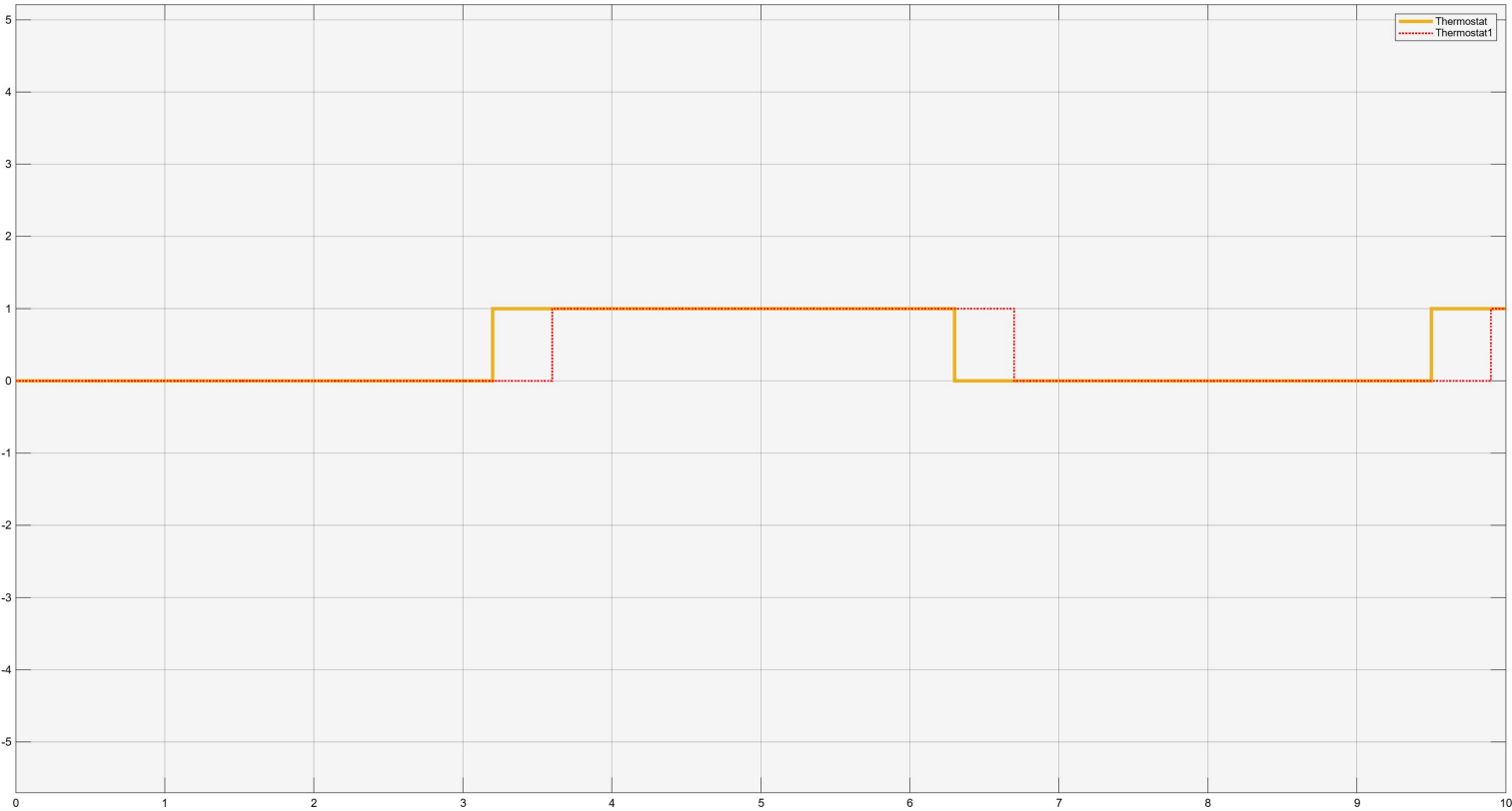


Figure 3.3: Scope plot comparing the system behaviour and mitigation of chattering.

Name: Srinjoy Chakraborty

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e.g. CSEHomework Assignment 1
Deadline: September 11, 2026**Problem 4.** (10 points)

Construct a timed automation model of a thermostat where the heater will be switched on when the room temperature is less than 22C and switched off when the temperature is above 26C. We also have the following additional requirement. Before switching on or off, it will generate a “beep” sound every second for 5 seconds as a warning. That means once the system senses that the temperature is less than 22C, it will generate the warning first, and then the heater will be switched on. Similarly, a warning sound will be issued before switching off the heater after the temperature reaches 26C.

I implemented the timed automaton using a Stateflow chart in Simulink. It has four main states that handle the temperature conditions as well as the 5-second warning before switching the heater:

- **Cooling:** Both the heater and beep are OFF. When `temperature < 22`, the system changes to `Warn_Heating`.
- **Warn_Heating:** The heater remains OFF during this stage. The `beep` signal is switched ON and OFF every 1 second. After 5 seconds, using `after(5, sec)`, the system moves to `Heating`.
- **Heating:** The heater is ON and the beep is OFF. When `temperature > 26`, the system changes to `Warn_Cooling`.
- **Warn_Cooling:** The heater stays ON in this state, while the `beep` keeps toggling every second for 5 seconds. After that, the system goes back to `Cooling`, where the heater is finally switched OFF.

I used a Sine Wave block with settings **Amplitude=5**, **Bias=24**, **Frequency=1** to simulate the room temperature, and routed both `heater` and `beep` outputs to a Scope to check that the timing and thresholds are working correctly.

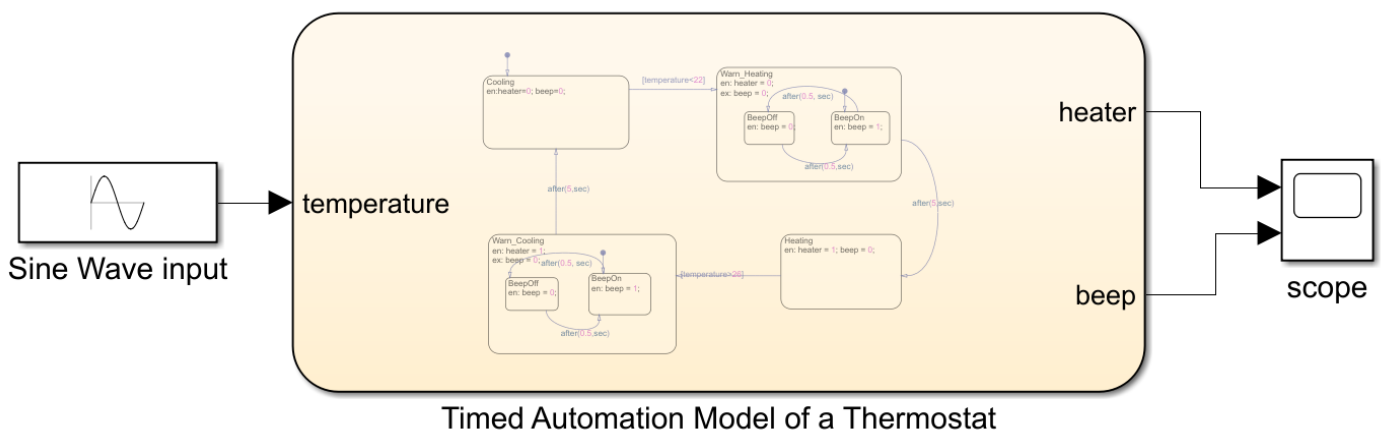


Figure 4.1: Main Simulink layout showing the input, Stateflow block, and Scope.

Name: Srinjoy Chakraborty

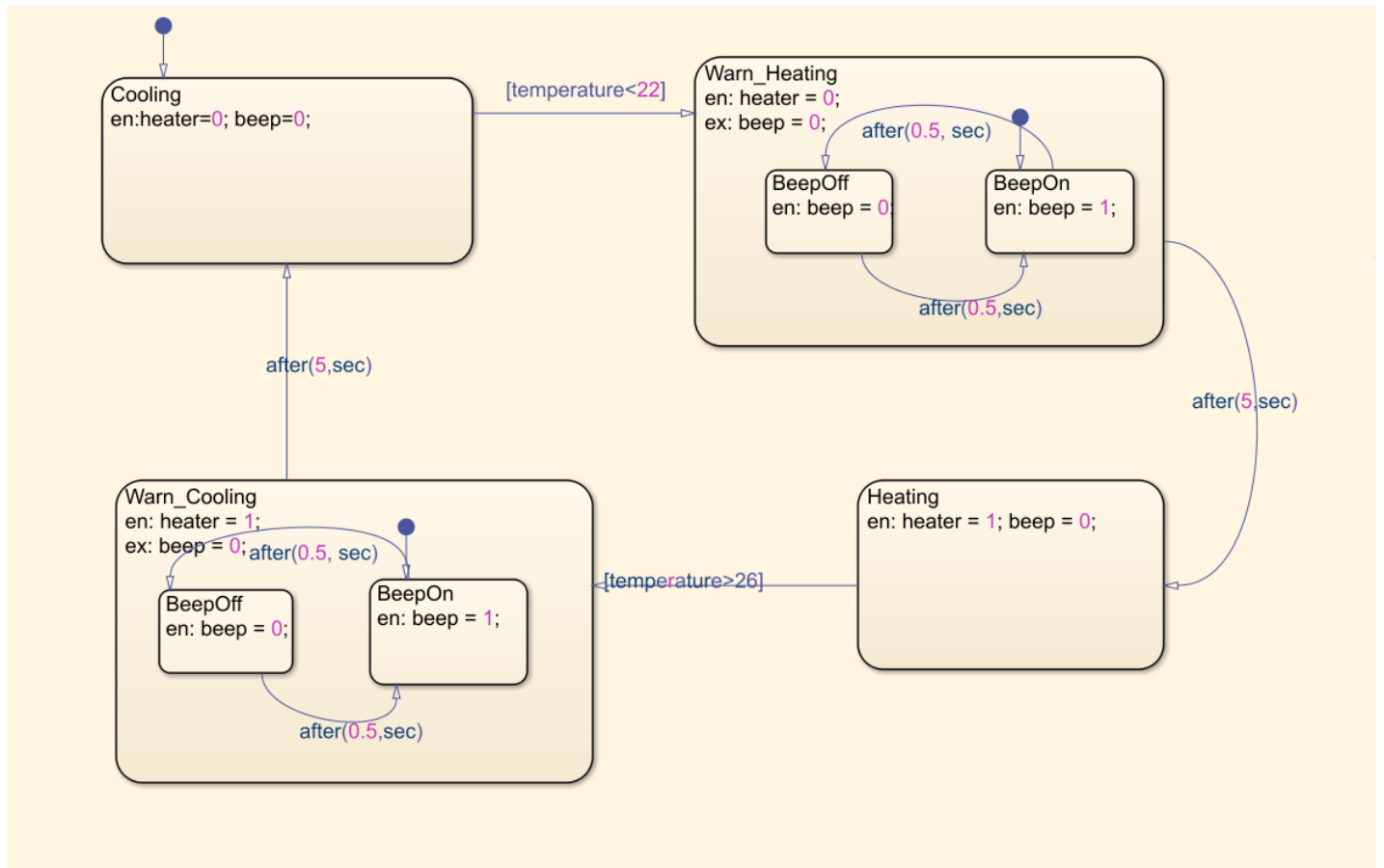
Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 4.2: Stateflow chart.

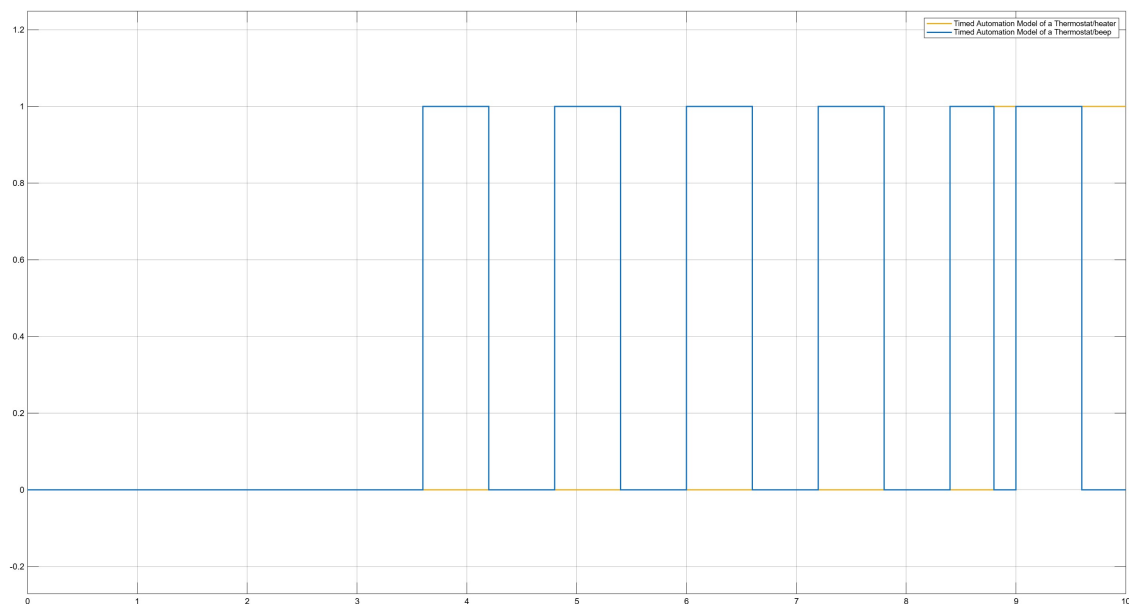
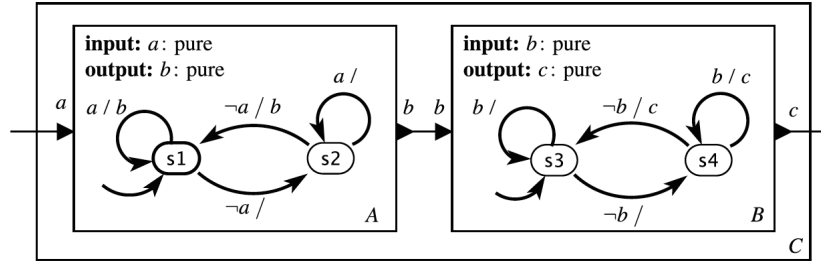


Figure 4.3: Scope plot.

Name: Srinjoy Chakraborty

Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026**Problem 5.** (10 points)

Consider the following synchronous composition of two state machines A and B.



Construct a single state machine C representing the composition. Which states of the composition are unreachable?

To build the combined machine C, the two state sets are combined to get the product state space

$$S_C = \{(s1, s3), (s1, s4), (s2, s3), (s2, s4)\}.$$

The initial state is (s1, s3). In the synchronous composition, Machine A takes the input a and produces the intermediate output b , which can either be present or absent. This output is then directly given to Machine B as its input, and Machine B produces the final output c .

Composition Transition Table:

Current State	Input a	Intermediate b	Next State	Output c
s1_s3	1	present	s1_s3	0
s1_s3	0	present	s2_s3	0
s1_s4	1	present	s1_s4	1
s1_s4	0	present	s2_s4	1
s2_s3	1	absent ($\neg b$)	s2_s4	1
s2_s3	0	absent ($\neg b$)	s1_s4	1
s2_s4	1	absent ($\neg b$)	s2_s3	0
s2_s4	0	absent ($\neg b$)	s1_s3	0

Reachability Analysis:

All four states are reachable, so there are no unreachable states. Starting from the initial state s1_s3, giving $a = 0$ moves the system to s2_s3. From s2_s3, an input of $a = 1$ moves it to s2_s4, while $a = 0$ moves it to s1_s4. In this way, all the states in the combined machine can be reached.

Name: Srinjoy Chakraborty

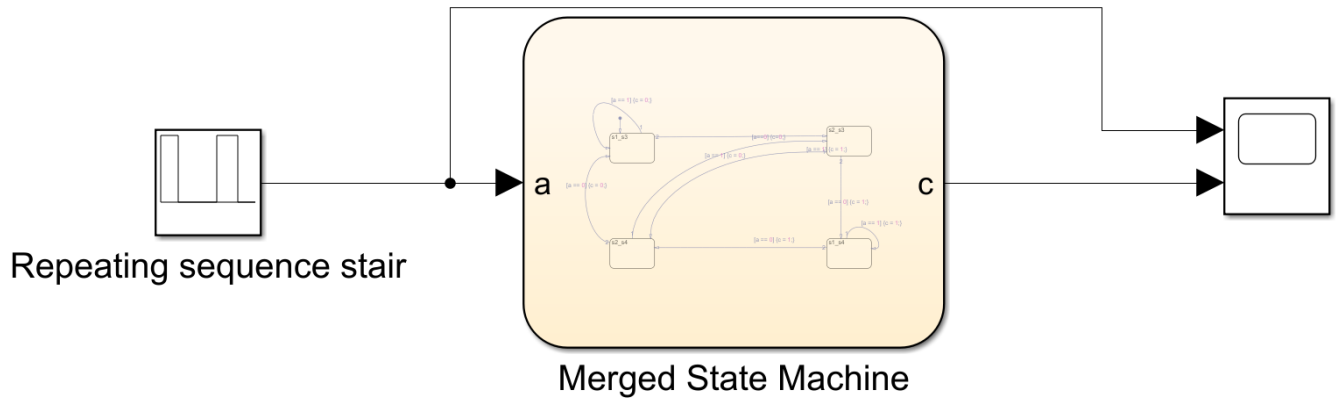
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e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 5.1: Simulink layout showing a Repeating sequence stair driving the Merged State Machine's input port 'a', with the signal 'a' and output port 'c' routed to a Scope for comparison.

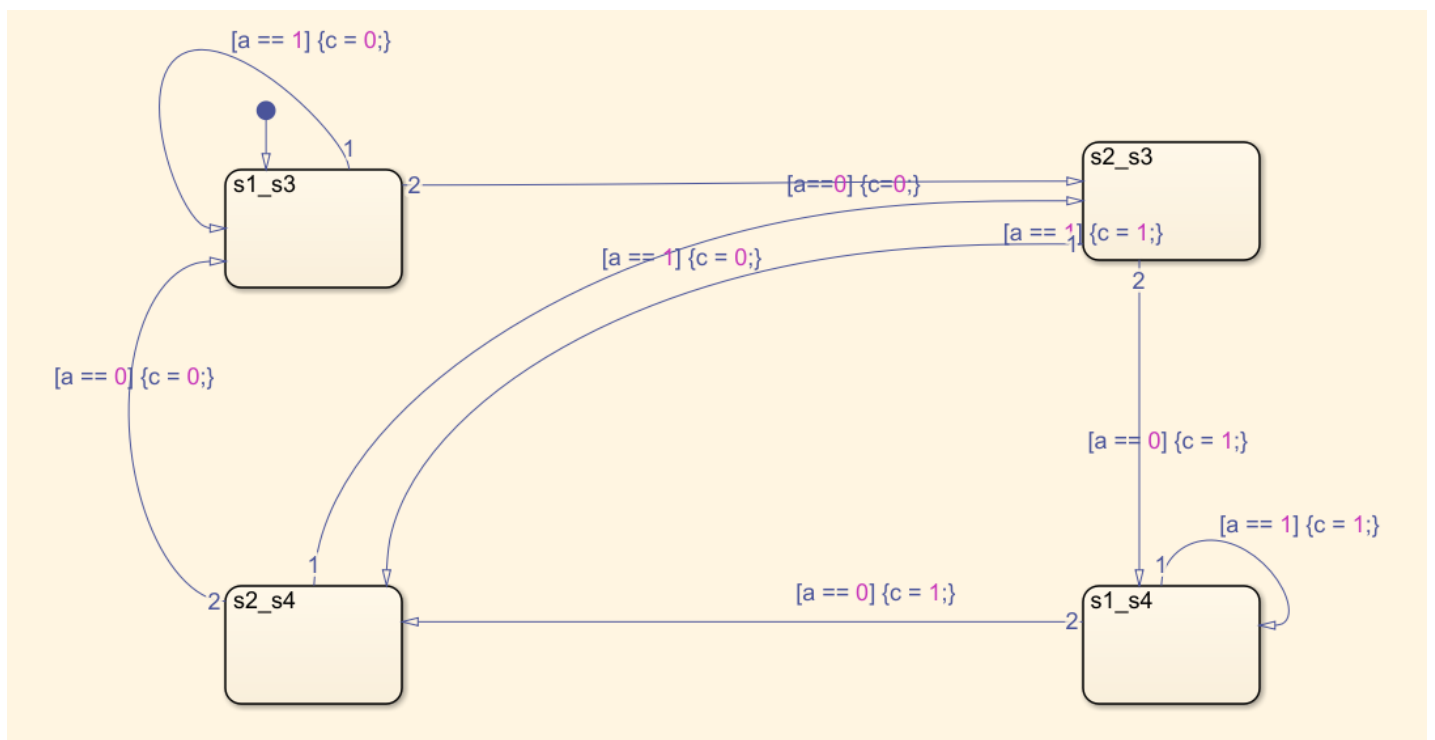


Figure 5.2: Stateflow chart.

Name: Srinjoy Chakraborty

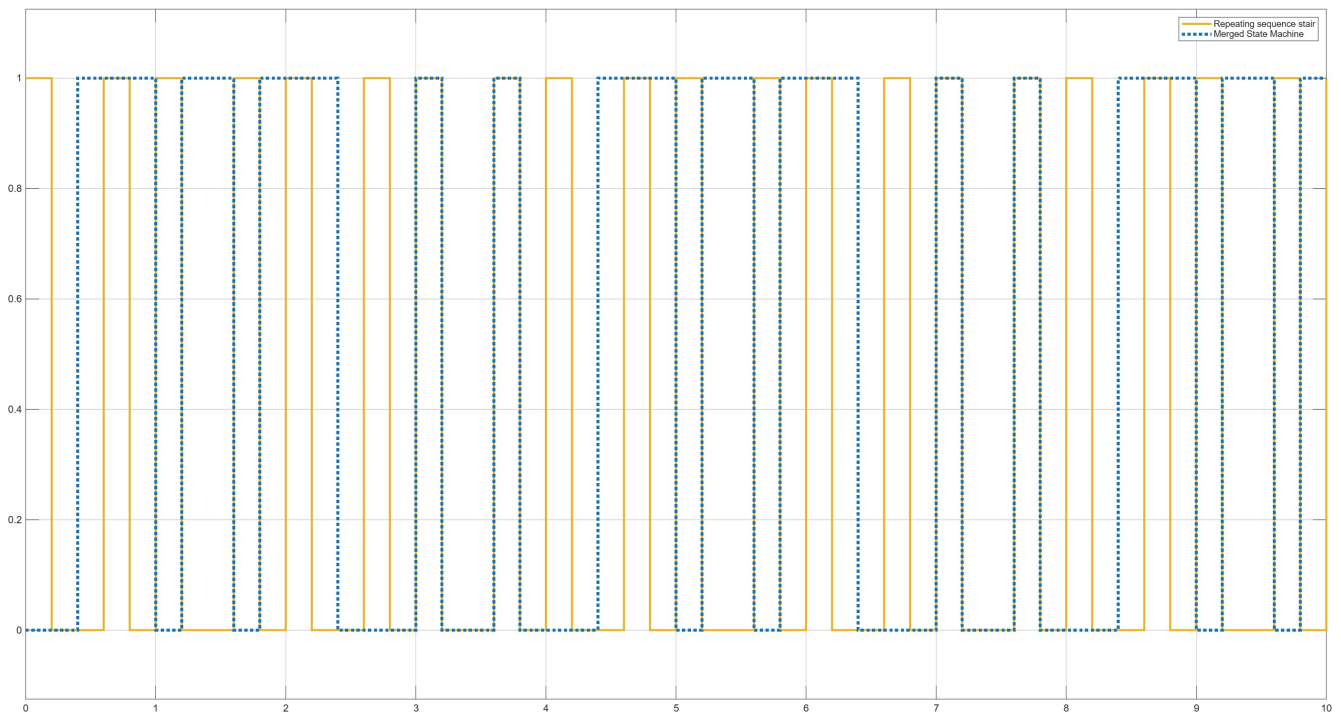
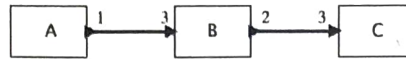
Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
e.g. 170001Dept.: EE
e.g. CSEHomework Assignment 1
Deadline: September 11, 2026

Figure 5.3: Scope plot demonstrating the dynamic output.

Name: SRINJOY CHAKRABORTY

Indian Institute of Technology Kanpur
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e.g. CSEHomework Assignment 1
Deadline: September 11, 2026**Problem 6.** (10 points)

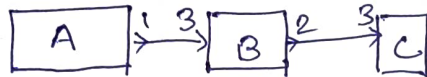
Consider the following SDF model:



The numbers adjacent to the ports indicate the number of tokens produced or consumed by the actor when it fires. Answer the following questions about this model.

(a) Let q_A , q_B , and q_C denote the number of firings of actors A , B , and C , respectively. Write down the balance equations and find the least positive integer solution.

(b) Find a schedule for an unbounded execution that minimizes the buffer sizes on the two communication channels. What is the resulting size of the buffers?

Figure

(a)

No. of firings are $\rightarrow q_A, q_B, q_C$

Edge A to B $\Rightarrow 1 \cdot q_A = 3 \cdot q_B$ - (i)

Edge B to C $\Rightarrow 2 \cdot q_B = 3 \cdot q_C$ - (ii)

Solving (i) & (ii)

$$q_A = 3q_B, \quad q_C = \frac{2}{3}q_B$$

The least possible solutions are:-

$$q_B = 3$$

$$\therefore q_A = 9, \quad q_C = 2$$

(b)

We need to minimize the buffer size.

$\therefore$ Fire the actor, the exact moment it has enough tokens.

Name: SRINJOY CHAKRABORTY

Roll No: 251040638

Dept.: EE
e.g. CSEIndian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

The overall token count change:-

① Fire A $\rightarrow$ 3 times. AB buffer = 3
BC buffer = 0

② B now has enough, fire B $\rightarrow$ 1 time.
AB = 0
BC = 2

③ As C needs 3 tokens, but has 2, fire A $\rightarrow$ 3 times.
B $\rightarrow$ 1 time.
 $\therefore$ AB = 0
BC = 4

④ Now C has enough. Here C is fired 1 time.
AB = 0
BC = 1

⑤ Again fire A $\rightarrow$ 3 times, B $\rightarrow$ 1 time
AB = 0
BC = 3

⑥ Fire C 1 time. AB = 0
BC = 0

Schedule $\rightarrow$ AAABAAABCAAAABC.Buffer SizeBuffer A $\rightarrow$ B : 3Buffer B $\rightarrow$ C : 4